

A Mathematical Tutorial on Ballistics for Precision Rifle Shooting

Contents

1	Introduction	2
2	Coordinate System and Notation	2
3	Basic Projectile Motion (Vacuum)	2
3.1	Limitations	2
4	Aerodynamic Drag	3
4.1	Drag Force Model	3
4.2	Ballistic Coefficient	3
5	1D Drag Model	3
6	Full 2D System with Drag	4
7	Numerical Integration	4
7.1	Euler Method	4
7.2	Runge-Kutta (RK4)	4
8	Wind Effects	4
8.1	Crosswind Drift	4
9	Spin Drift and Coriolis Effect	5
9.1	Spin Drift	5
9.2	Coriolis Effect	5
10	Trajectory Correction	5
10.1	Drop Compensation	5
10.2	Minute of Angle (MOA)	5
10.3	Milliradians (MIL)	5
11	Atmospheric Effects	6
11.1	Air Density	6
12	Practical Ballistic Modeling	6
12.1	G1 and G7 Models	6
12.2	Table-Based Integration	6
13	Example Calculation	6

1 Introduction

Precision rifle shooting relies on understanding the physics governing projectile motion. Unlike idealized textbook problems, real-world ballistics involves:

- Gravity
- Aerodynamic drag
- Wind effects
- Bullet stability and spin

This tutorial develops the mathematical framework necessary to model bullet trajectories accurately.

2 Coordinate System and Notation

We define:

- x : horizontal distance (downrange)
- y : vertical height
- z : lateral deviation (wind direction)

Velocity vector:

$$\mathbf{v} = (v_x, v_y, v_z)$$

Acceleration:

$$\mathbf{a} = \frac{d\mathbf{v}}{dt}$$

3 Basic Projectile Motion (Vacuum)

Ignoring air resistance:

$$\mathbf{a} = (0, -g, 0)$$

Solution:

$$x(t) = v_0 \cos \theta \cdot t \tag{1}$$

$$y(t) = v_0 \sin \theta \cdot t - \frac{1}{2}gt^2 \tag{2}$$

Eliminating t :

$$y(x) = x \tan \theta - \frac{gx^2}{2v_0^2 \cos^2 \theta}$$

3.1 Limitations

This model is insufficient because:

- Bullets travel at supersonic speeds
- Drag dominates trajectory behavior

4 Aerodynamic Drag

4.1 Drag Force Model

Drag force is:

$$\mathbf{F}_D = -\frac{1}{2}\rho C_D A v \mathbf{v}$$

where:

- ρ : air density
- C_D : drag coefficient
- A : cross-sectional area
- $v = \|\mathbf{v}\|$

Equation of motion:

$$m \frac{d\mathbf{v}}{dt} = -\frac{1}{2}\rho C_D A v \mathbf{v} + (0, -mg, 0)$$

4.2 Ballistic Coefficient

Define ballistic coefficient:

$$BC = \frac{m}{C_D A}$$

Rewriting:

$$\frac{d\mathbf{v}}{dt} = -\frac{\rho}{2BC} v \mathbf{v} + (0, -g, 0)$$

5 1D Drag Model

Consider horizontal motion only:

$$\frac{dv}{dt} = -kv^2, \quad k = \frac{\rho}{2BC}$$

Separable equation:

$$\int \frac{dv}{v^2} = -k \int dt$$

Solution:

$$-\frac{1}{v} = -kt + C$$

$$v(t) = \frac{v_0}{1 + kv_0 t}$$

Position:

$$x(t) = \frac{1}{k} \ln(1 + kv_0 t)$$

6 Full 2D System with Drag

We solve:

$$\frac{dv_x}{dt} = -kvv_x \quad (3)$$

$$\frac{dv_y}{dt} = -kvv_y - g \quad (4)$$

with:

$$v = \sqrt{v_x^2 + v_y^2}$$

This system has no closed-form solution and must be solved numerically.

7 Numerical Integration

7.1 Euler Method

$$v_x^{n+1} = v_x^n + \Delta t(-kv^n v_x^n) \quad (5)$$

$$v_y^{n+1} = v_y^n + \Delta t(-kv^n v_y^n - g) \quad (6)$$

7.2 Runge-Kutta (RK4)

More accurate:

$$\mathbf{v}_{n+1} = \mathbf{v}_n + \frac{\Delta t}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Used in modern ballistic solvers.

8 Wind Effects

Assume wind velocity $\mathbf{w}$.

Relative velocity:

$$\mathbf{v}_{rel} = \mathbf{v} - \mathbf{w}$$

Drag becomes:

$$\mathbf{F}_D = -\frac{1}{2}\rho C_D A \|\mathbf{v}_{rel}\| \mathbf{v}_{rel}$$

8.1 Crosswind Drift

Approximate lateral drift:

$$z \approx \frac{w}{v_{avg}} x$$

More accurate models require full 3D integration.

9 Spin Drift and Coriolis Effect

9.1 Spin Drift

Due to gyroscopic stability:

$$z_{spin} \propto x^2$$

Typically small but significant at long range.

9.2 Coriolis Effect

Earth rotation introduces acceleration:

$$\mathbf{a}_c = -2\mathbf{\Omega} \times \mathbf{v}$$

Magnitude:

$$a_c \approx 2\Omega v$$

where $\Omega \approx 7.29 \times 10^{-5}$ rad/s.

10 Trajectory Correction

10.1 Drop Compensation

Vertical drop:

$$\Delta y \approx \frac{1}{2}gt^2$$

Converted to angular correction:

$$\theta \approx \frac{\Delta y}{x}$$

10.2 Minute of Angle (MOA)

1 MOA $\approx$ 1.047 inches at 100 yards

Correction:

$$\text{MOA} = \frac{\Delta y}{x} \cdot \frac{180 \times 60}{\pi}$$

10.3 Milliradians (MIL)

1 mil = 10^{-3} rad

$$\text{mil} = \frac{\Delta y}{x} \times 1000$$

11 Atmospheric Effects

11.1 Air Density

$$\rho = \frac{P}{RT}$$

where:

- P : pressure
- T : temperature
- R : gas constant

Higher altitude $\Rightarrow$ lower $\rho \Rightarrow$ less drag.

12 Practical Ballistic Modeling

12.1 G1 and G7 Models

Drag coefficient depends on velocity:

$$C_D = f(v)$$

Standard drag models:

- G1: flat-base bullets
- G7: boat-tail bullets

12.2 Table-Based Integration

Modern solvers use:

- Tabulated drag curves
- Interpolation
- Numerical integration

13 Example Calculation

Given:

- $v_0 = 800$ m/s
- $BC = 0.5$
- Distance = 500 m

Estimate time of flight (approximate):

$$t \approx \frac{x}{v_{avg}} \approx \frac{500}{700} \approx 0.71 \text{ s}$$

Drop:

$$\Delta y = \frac{1}{2}gt^2 \approx 2.5 \text{ m}$$

MIL correction:

$$\text{mil} = \frac{2.5}{500} \times 1000 = 5 \text{ mil}$$

14 Conclusion

Precision shooting requires combining:

- Physics (drag, gravity)
- Mathematics (ODEs, numerical methods)
- Environmental corrections

Modern ballistic solvers implement these equations numerically for high accuracy.